Sustainable Energy Patterns Across Countries (2000–2020)

This project explores how energy and economic indicators relate to each other across 159 countries from 2000 to 2020. The analysis is based on **SDG 7: “Affordable and Clean Energy”**, and uses the Global Data on Sustainable Energy dataset from Kaggle as the main source.

Dataset and Variables

The raw dataset contains 3,649 rows and 21 columns which are related to energy, environmental, and economic for 176 different countries from 2000 to 2020. After dropping columns with more than 50% missing values and columns which are not relevant to the analysis, we kept 8 columns:

Entity: country name
Year: year of observation (2000–2020)
Access to Electricity: % of population with electricity access
Renewable Share: % of renewable energy in total final energy consumption
Energy Intensity: energy used per unit of GDP (MJ/$2017 PPP GDP)
Energy per Capita: primary energy consumption per person (kWh/person)
GDP Growth: yearly GDP growth rate (%)
GDP per Capita: GDP per person (USD)

Methodology

Missing values were filled using three different methods depending on the variable: Linear Regression for variables with clear trends (Access to Electricity and GDP per Capita), Linear Interpolation for Energy Intensity (applied only to countries with have a large amount notna data), and Median per country for GDP Growth, which is right-skewed. After cleaning, the dataset has 3,325 rows with no missing values. We then produced 5 visualizations: a correlation heatmap, a bar chart of renewable share by access level, a scatter plot of wealth vs energy consumption, a line plot of efficiency trends by income group, and 2 side bar charts of increasing and decreasing countries in renewable share change.

Key Findings

1. Renewable share decreases as electricity access increases (r = -0.79). Countries with low electricity access (<25%) have an average renewable share of 80.9%, while countries with full access (up to 100%) only have 14.6%. The pattern is linear, the higher the access to electricity, the lower the renewable share.
2. Energy consumption highly correlates with wealth (r = +0.66). The richest 10% of countries use around 115 times more energy per person than the poorest 10%. On a logarithmic scale, the relationship is multiplicative meaning that every time the wealth doubles, the energy consumed also doubles.
3. All four income groups reduced their energy intensity between 2000 and 2020 which means that the world is becoming more efficient at using energy to improve their

economy. The biggest improvement came from Low Income countries where they reduce their energy intensity 24%. Meanwhile, Lower-Middle Income countries improved the least by reducing only 6.9%.

4. The biggest renewable share increase from 2000 to 2020 came from Iceland (+14.7%), Zimbabwe (+12.6%), and Gabon (+11.7%). The biggest decreases came from Equatorial Guinea (-39.1%), Afghanistan (-25.4%), and Ghana (-19.7%). Looking at the increasing and decreasing country in renewable shares together, there is no single pattern based on wealth alone. Both rich countries (Iceland, Denmark, Germany) and poor countries (Zimbabwe, Eswatini) appear in the increasing list. Meanwhile, most decreasing ones have low to medium GDP per capita, but Equatorial Guinea (the biggest decreasing) actually has a higher GDP than several increasing countries.

Conclusions

The four findings above show one main lesson: we should not look at energy numbers one by one. A single number, like renewable share or energy intensity, can mean very different things depending on the country. For example, two countries can have the same renewable share but be in very different situations. To really understand if a country is making progress in sustainability, we need to look at several numbers together and also think about the country's situation, such as how developed it is.

Limitations

- Most imputations in this dataset were filled in using statistical methods (Linear Regression, Interpolation, and Median). This means that the filled values are estimates, not real measurements.
- The dataset's "Renewable Share" column does not separate the types of renewable energy. Different sources such as solar, wind, hydro, geothermal, and traditional renewable energy(firewood, charcoal, animal waste) are all combined into a single percentage. This means a high renewable share could come from modern clean energy or from traditional fuel use, and we cannot tell the difference from this data alone.
- All findings are correlations but not all findings are causation relationships. We can say two variables are correlated with each other, but we can’t say for all variables that it causes the other variables.

Data Source

- Ansh Tanwar. Global Data on Sustainable Energy (2000-2020). Kaggle. https://www.kaggle.com/datasets/anshtanwar/global-data-on-sustainable-energy
- Income classification thresholds from World Bank Data Help Desk: https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups

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